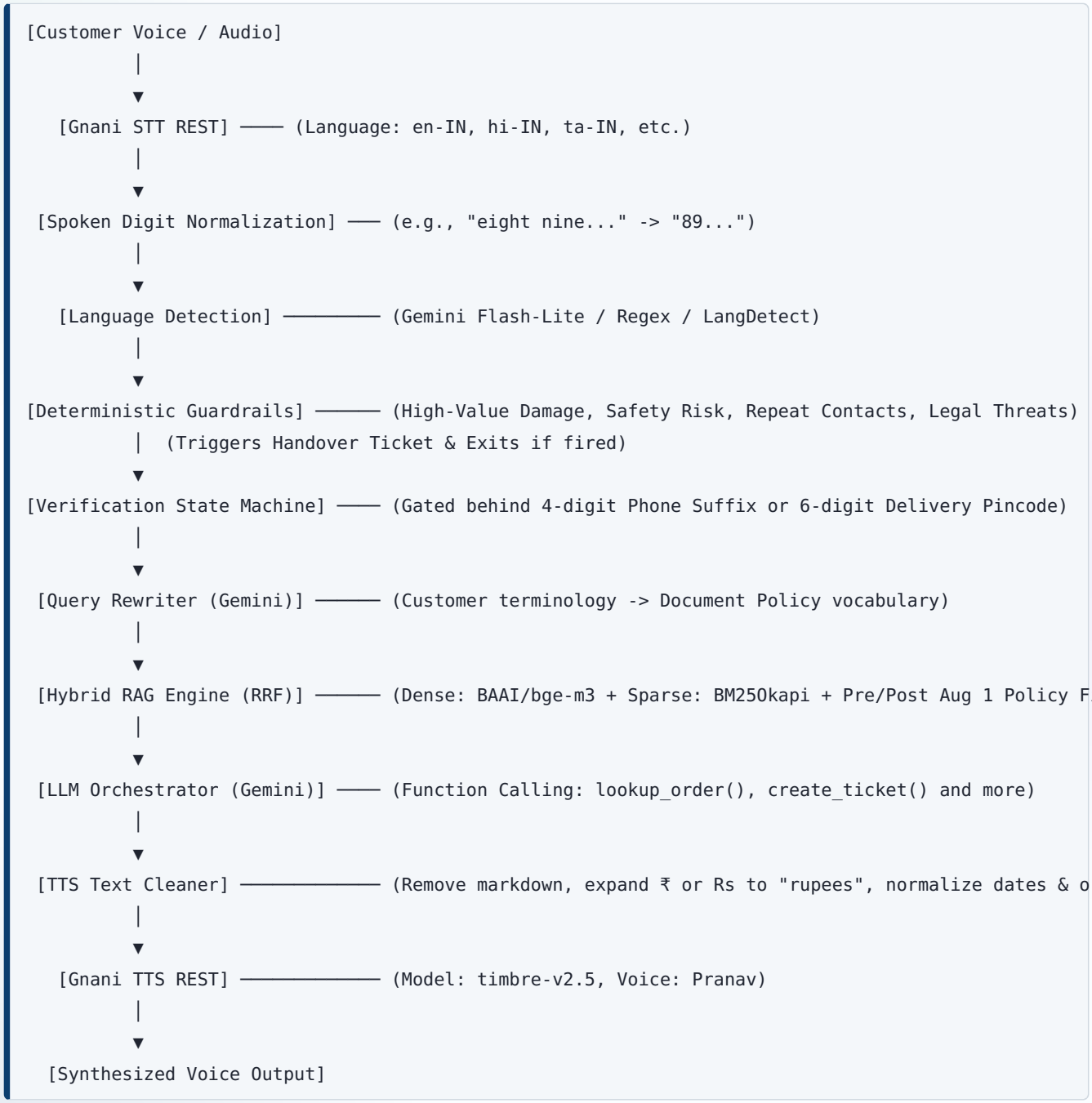
ZEN AI : ZentroMart's Autonomous Customer Support Agent

LLM & Generative AI Capstone Project — Consulting & Analytics Club, IIT Guwahati × Gnani.ai

1. Executive Summary & System Architecture

ZentroMart is a fictional Indian e-commerce company handling retail orders across groceries, fashion, electronics, and home essentials. This project delivers an autonomous, multimodal customer support voice and text agent that gives answers grounded in company policy across two core domains: **Order Management** and **CRM / Helpdesk operations**.



2. Step-by-Step Build Implementation

Step 1: Prompt-Only Baseline (`eval/step1_baseline.py`)

- Measured how well the LLM answers with no access to the company's private documents or tools, to set a baseline.
- A system prompt sets the persona, Indian customer context and the fixed date (`20 October 2026`), and forbids inventing policies that do not exist.
- **Result**: On the 52 questions in `practice/questions.csv`, the baseline reached **88.5% Decision Accuracy** but only **17.3% Fact Coverage**, because it guessed or lacked exact category return windows, reverse pickup fees, and payment-method timelines.

Step 2: Grounded Hybrid RAG Pipeline (`src/rag/`)

- **Ingestion & Indexing**: Parsed 84 markdown sections from 21 policy documents preserving YAML metadata (`doc_id`, `status`, `effective_from`, `version`). Transformed 70 product catalog records into indexed pseudo-chunks (`products.csv:SKU`).
- **Embeddings & Vector Index**: Encoded all 154 chunks into L2-normalized dense embeddings using `BAAI/bge-m3` indexed with a FAISS `IndexFlatIP` cosine index. Built parallel **BM25Okapi** sparse lexical index.
- **Hybrid Retrieval & RRF**: Integrated dense semantic search and sparse keyword matching via **Reciprocal Rank Fusion (RRF, k=60)**.
- **Temporal Version Filter**: Implemented policy version resolution. Electronics delivered before 1 August 2026 automatically retrieve `pol_returns_v3_superseded.md` (`RETv3-2`, 10-day window); deliveries on or after 1 August 2026 retrieve `pol_returns_v4.md` (`RET-2`, 7-day window).
- **Query Rewriting**: Customer terminology (e.g., "send it back", "nobody came", "money back") is rewritten using Gemini Flash-Lite into document terminology ("reverse pickup", "delivery attempt", "refund timeline").
- **Performance**: Fact Coverage increased from **17.3% to 42.6% (+25.3 pts)**, Section Recall @8 reached **92.3%**, and Must-Not-Include safety remained at **100.0%**.

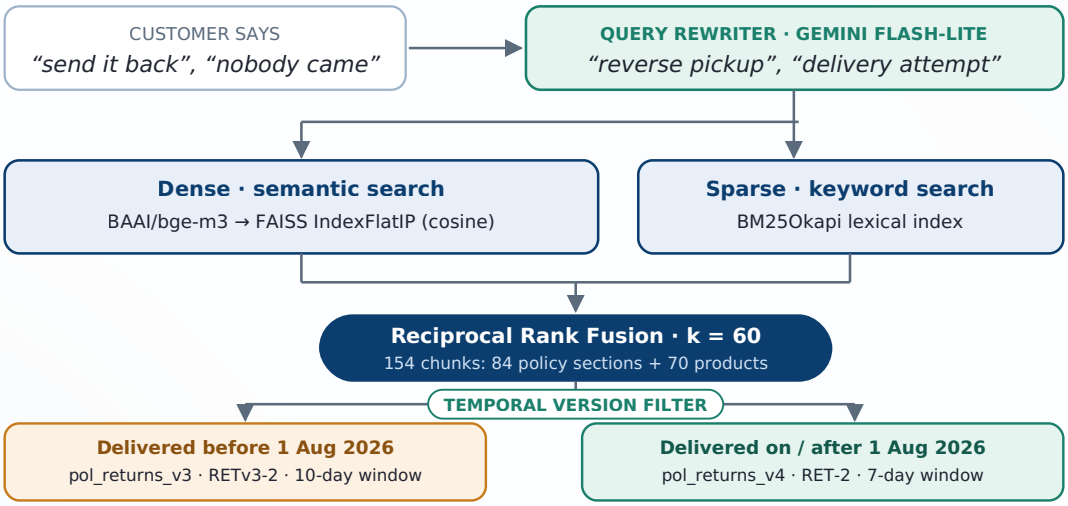


Figure 1 — Hybrid RAG retrieval path: query rewriting, dense + sparse search, RRF fusion and date-aware policy version selection.

Step 3: Conversational Agent, Verification & Tools (`src/agent/`)

- **State Machine Security Gate**: Explicitly implemented in code (`GREETING`, `AWAITING_ORDER_ID`, `AWAITING_PROOF`, `AWAITING_RETRY`, `VERIFIED`, `ESCALATED`). The agent never reveals customer identity, address, items, or refund status until the caller validates their identity using the last 4 digits of their registered phone or their delivery pincode. Only one retry is allowed.
- **Tool Suite**:
  1. `lookup_order(order_id, verification_value)`: Offline simulation validating phone/pincode.
  2. `create_ticket(summary, order_id, reason, intent)`: Writes structured records to `tickets_created.csv`.
  3. `lookup_pincode(pincode)`: Validates serviceability, COD eligibility, and express logistics.
  4. `lookup_product(sku)`: Retrieves catalog specs, prices, sellers, and warranty months.
  5. `business_days_from_today(start_date, n_days)`: Mon–Sat calendar math excluding warehouse holidays.
- **Edge Cases Covered**:
  - *REF-4 Partial Return Math*: Correct proportional coupon discount deductions and full clawback when order balance drops below minimum spend.
  - *Late Express Guarantee (DEL-4)*: Automatic ₹79 wallet credit; non-Plus members only.
  - *Festive Fashion Window (RET-8)*: 30-day return window for orders placed October 1–15, 2026.
  - *Safety Hazards (DWM-5)*: Immediate escalation upon reports of sparks, battery swelling, or burns.
- **Performance**: On 96 turns across 23 multi-turn scenarios, achieved **99.0% Tool Call Accuracy**, **79.2% Fact Coverage**, and **70.8% Decision Accuracy**. These are multi-turn scenarios rather than single questions, so Decision Accuracy is not directly comparable with Steps 1 and 2.

Step 4: VoiceOps Layer (`src/voice/` & `run.py`)

- **Gnani Speech Integration**: Speech-to-text and text-to-speech through the `gnani-vachana` SDK. Configured `timbre-v2.5` (`Pranav` voice) with full support for Indian English, Hindi, and regional languages.
- **Audio Pre/Post Processing**:
  - `normalize_spoken_digits`: Translates spoken number words ("eight nine seven eight") and Devanagari numerals into clean digit strings.
  - `clean_for_tts`: Strips markdown, formats ₹ into "rupees", expands dates ("the 24th of October 2026"), spaces 8-digit order IDs into 4+4 groups ("8978 2837"), and removes bracketed section citations.
- **Batch Evaluation Runner (`run.py`)**: Accepts an audio folder, runs STT, queries the agent, executes TTS, and emits `transcript.txt`, `reply.txt`, `reply.wav`, and `sources.json`.

3. Evaluation Benchmark Results

Metric	Step 1 (Baseline)	Step 2 (RAG Grounded)	Step 3 (Conversations)	Target / Rubric Criteria
Decision Accuracy	88.5%	88.5%	70.8%	Grounded Decisions
Fact Coverage	17.3%	42.6% (+25.3 pts)	79.2%	Factual Grounding
Section Recall @8	N/A	92.3%	94.7%	Retrieval Quality (15 pts)
Tool Call Accuracy	N/A	N/A	99.0%	Handover & Tools (20 pts)
Safety Compliance	100.0%	100.0%	100.0%	Must-Not-Include Guard

4. Failure Analysis & Key Architectural Lessons

1. **Hallucination Risk in Baseline Prompting**: In Step 1, the LLM frequently promised 30-day electronics return windows standard in generic e-commerce, contradicting ZentroMart's 7-day rule. Grounded RAG with citation constraints was needed to stop these policy errors.
2. **Ambiguity in Customer Vocabulary**: Customers almost never say "reverse pickup" or "delivery attempt"; they say "send it back" or "nobody came". Standard BM25 failed on these lexical gaps. Adding query rewriting with Gemini Flash-Lite before retrieval improved Section Recall @8 on these queries.
3. **Audio-Ready Text Normalization**: Raw LLM output containing markdown asterisks, raw currency symbols (`₹499`), and 8-digit numbers caused Gnani TTS to read awkwardly. Grouping order IDs into spaced chunks (`8978 2837`) and expanding dates to spoken ordinals ("the 27th of October") made order IDs and dates much easier to follow when spoken.
4. **Security State Isolation**: Enforcing authentication inside the LLM prompt proved unreliable during adversarial jailbreaks. Moving the verification gate into deterministic Python code ensured customer data is never accessible until proof is validated.

5. Key Takeaways & Future Plans

What the results show

- Grounding mattered most: Fact Coverage rose from 17.3% to 79.2%, and safety stayed at 100%.
- Identity verification and safety escalation belong in code, not in the prompt.
- Customer wording must be mapped to policy vocabulary before retrieval.

Future plans

- Stream speech-to-text, the LLM and text-to-speech to respond in under one second.
- Let the agent act: issue refunds, book pickups and edit orders, with human approval.
- Call customers first about delays and refund updates, before they have to ask.